COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 2 : Session – 7 : SLO – 1 | Representation of decimal numbers in BCD format
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Match the Following (Answers)

- A. 9 → 4. 1001
- B. 15 → 1. 0001 0101
- C. 28 → 5. 0010 1000
- D. 76 → 3. 0111 0110
- E. 100 → 2. 0001 0000 0000

Fill in the Blanks

- In BCD, each decimal digit is represented using 4 bits.
- The BCD of the decimal number 12 is 0001 0010.
- BCD stands for BINARY CODED DECIMAL.
- In BCD, the decimal number 99 is written as 1001 1001.
- BCD encoding is mainly used in DIGITAL DISPLAY devices like calculators and digital clocks.

Activity Task – Convert and Create

- Birthdate (DDMM, sample 1508): 1 → 0001, 5 → 0101, 0 → 0000, 8 → 1000.
- BCD of 1508 = 0001 0101 0000 1000.
- Method: convert each decimal digit separately into its 4-bit binary equivalent and concatenate the groups — this is what distinguishes BCD from straight binary (1508 in pure binary would be 10111100100).